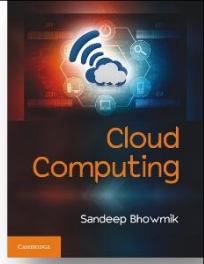
Cloud Computing

Sandeep Bhowmik

Chapter 5

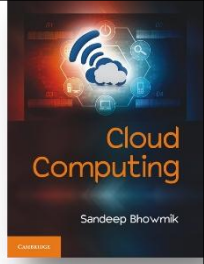
Cloud Computing Services

Cambridge University Press



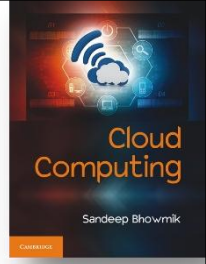
Cloud Computing Services

- There are three categories of computing services those people consume -
  - Infrastructure Service
  - Platform Service
  - Software Application Service
- Cloud computing talks about delivering these facilities to consumers as computing services through network/Internet.



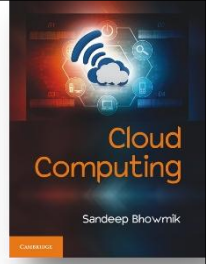
Cloud Computing Services

- Cloud computing offers computing infrastructure, platform and application, delivered 'as-a-service'.
- These services are referred as –
  - Infrastructure as a Service (IaaS)
  - Platform as a Service (PaaS)
  - Software as a Service (SaaS)



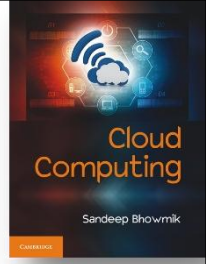
Cloud Computing Services

- IaaS, PaaS and SaaS refer to the delivery of computing services to consumers over network.
- These three service models together is commonly referred as SPI (for Service-Platform-Infrastructure) model.
- Cloud service providers arrange these services for the cloud consumers.



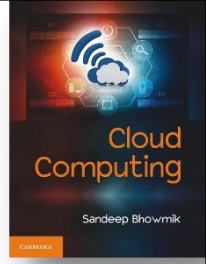
Infrastructure as a Service (IaaS)

- Hardware resources in cloud computing are not directly accessible to users.
- The hardware resources are represented as simulated software components.
- These recourses are known as virtual resources or virtual components.



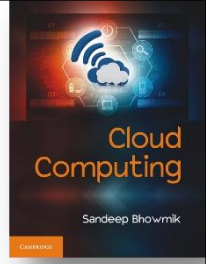
Infrastructure as a Service (IaaS)

- IaaS model provides all hardware components in virtual mode.
- It provides the facility of using virtual processor, memory, storage and network resources to the consumers.
- IaaS delivers virtualized-hardware resources to consumers via network or internet.



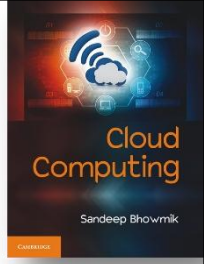
Infrastructure as a Service (IaaS)

- Consumers can use the virtual components just like real hardware devices.
- IaaS is also referred as Hardware as a Service (HaaS).
- The consumers no more require managing the underlying computing infrastructure that they consume as IaaS.
- The responsibility of managing the underlying hardware are outsourced to cloud service provider.



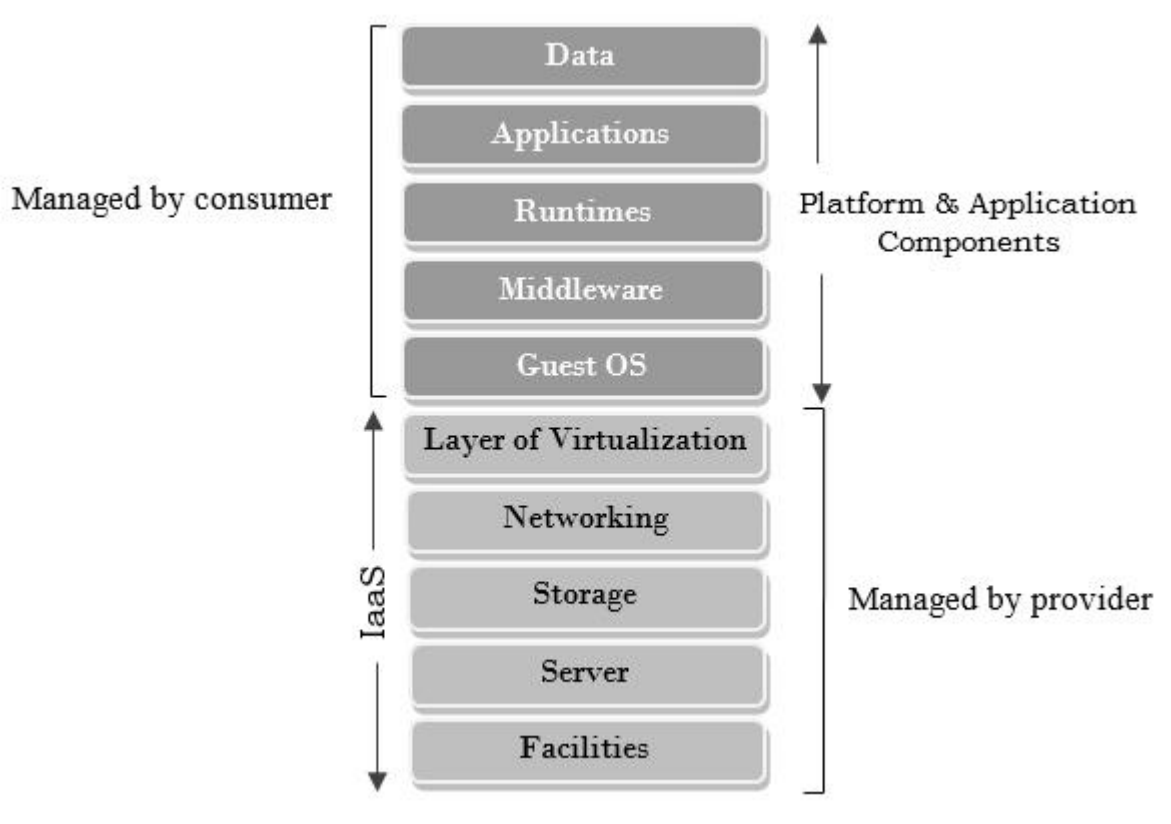
Infrastructure as a Service (IaaS)

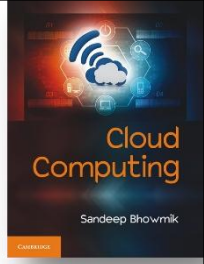
- Consumers can access these virtual hardware resources, on-demand, anytime from any location over the network.
- They can build computers (virtual computers) using those virtual (or virtualized) hardware components.
- Major computing vendors like Amazon, Google, GoGrid, RackSpace provide IaaS facility.



Infrastructure as a Service (IaaS)

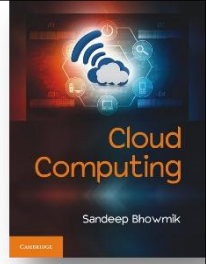
- **IaaS component stack**





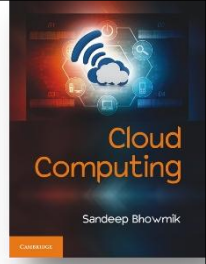
Infrastructure as a Service (IaaS)

- Cloud vendors deliver customized and readymade virtual machines via IaaS offering.
- Example of popular server environments -
  - Amazon EC2
  - Google Compute Engine



Platform as a Service (PaaS)

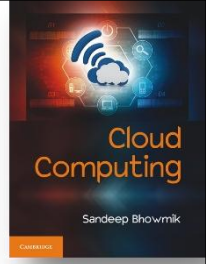
- A computing platform comprises of the hardware resources, operating system, middleware (if required) and runtime libraries.
- Platform means the underlying system on which software applications can be installed (and also developed).



Platform as a Service (PaaS)

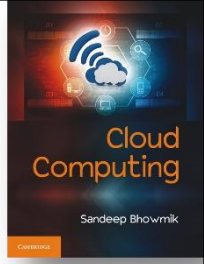
- Application development and deployment in traditional computing requires users' participation in managing hardware, operating system, middleware, web servers, and other components.

- PaaS facility on the other hand, relives users from all these headaches.



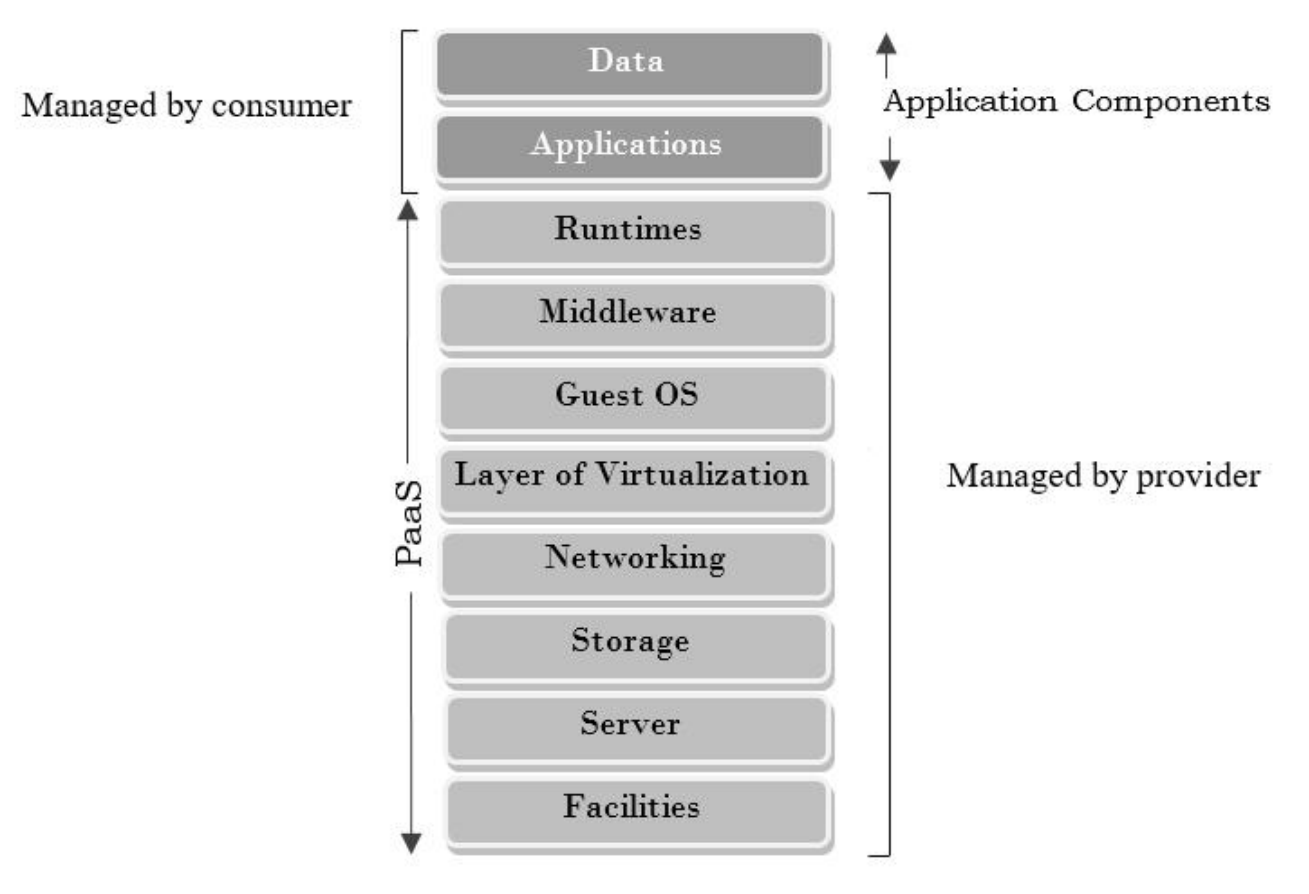
Platform as a Service (PaaS)

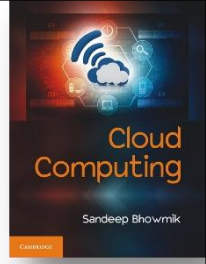
- PaaS component stack additionally provides application (development and deployment) platform over IaaS component stack.
- Thus, PaaS is created by adding additional layers of software over IaaS.
- PaaS model lets the users focus only on development and deployment of application, without the headache of arranging and managing the underlying hardware and software.



Platform as a Service (PaaS)

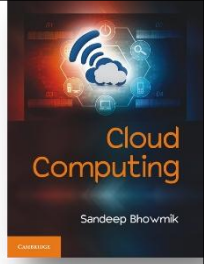
- **PaaS component stack**





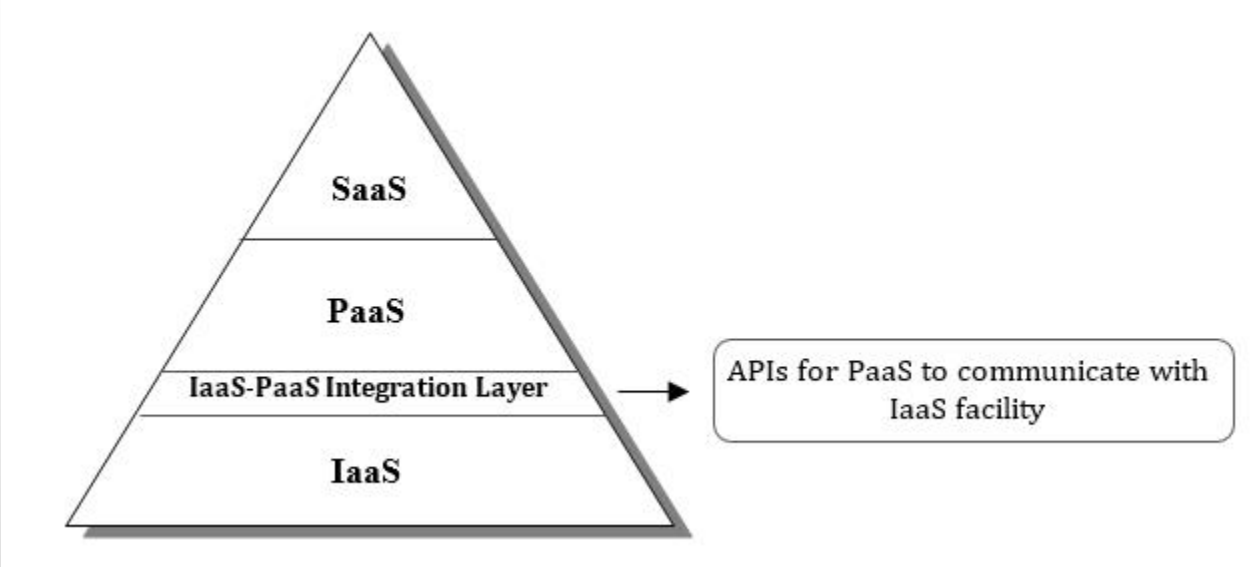
Platform as a Service (PaaS)

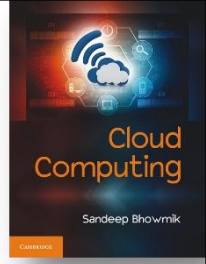
- Popular PaaS offerings –
  - Google App Engine
  - Microsoft Azure Platform
  - GoGrid CloudCenter
  - Force.com
- Thus, PaaS is created by adding additional layers of software over IaaS.



Platform as a Service (PaaS)

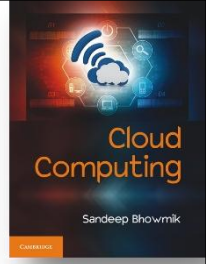
- **PaaS-IaaS Integration**





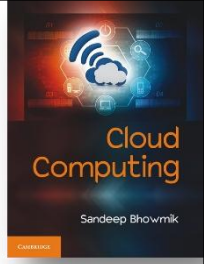
Software as a Service (SaaS)

- SaaS is a way of delivering application as-a-service, over the Internet/network.
- Users can directly use the software without the headache of installing or configuring an application.
- It also relieve consumers from the responsibility of arranging the infrastructure and platform.



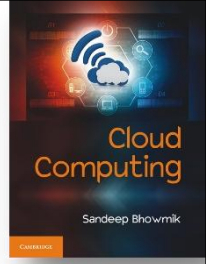
Software as a Service (SaaS)

- Customers do not need to buy software licenses or any additional computing resources to support application.
- Users can access applications against some rental fee on use basis.
- SaaS applications are sometimes referred as *web-based software*, or *hosted software*.



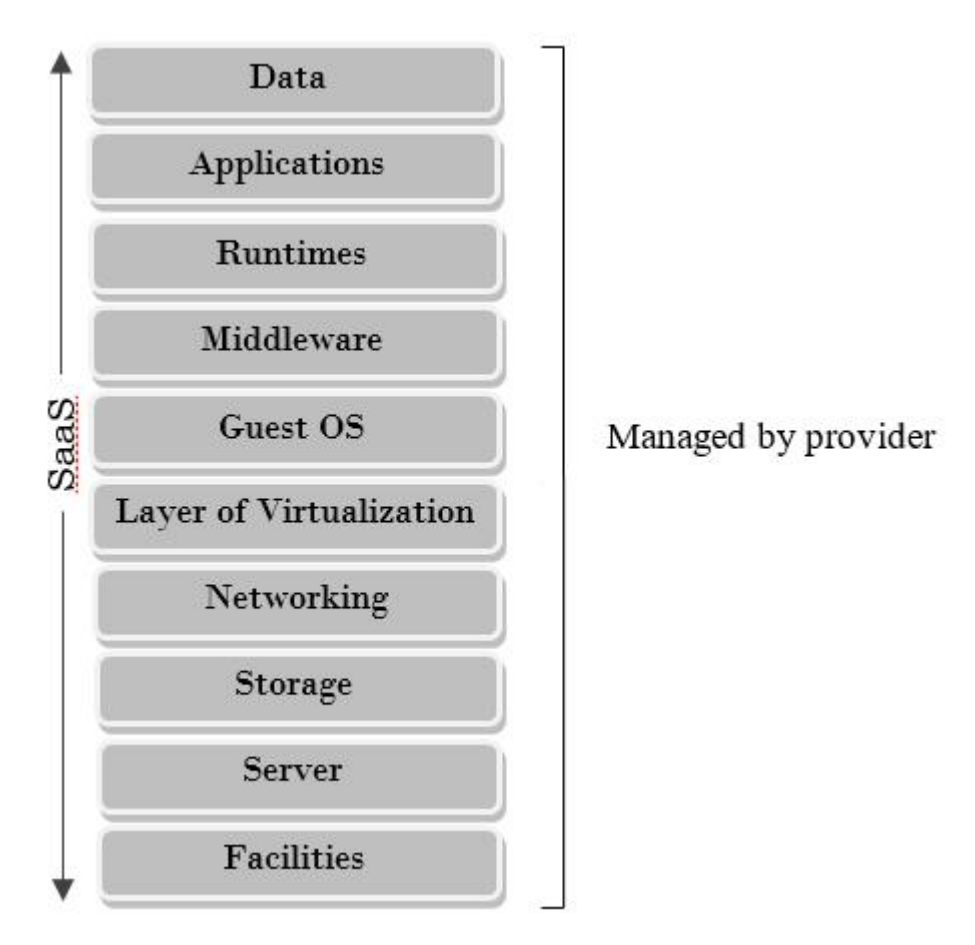
Software as a Service (SaaS)

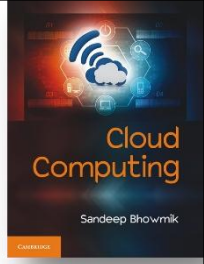
- In SaaS model, everything is managed by vendor including application upgrade or updates.
- SaaS is built by adding layers over PaaS component stack .



Software as a Service (SaaS)

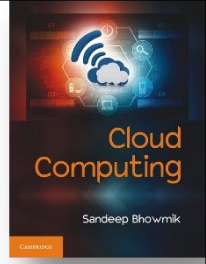
- **SaaS component stack**





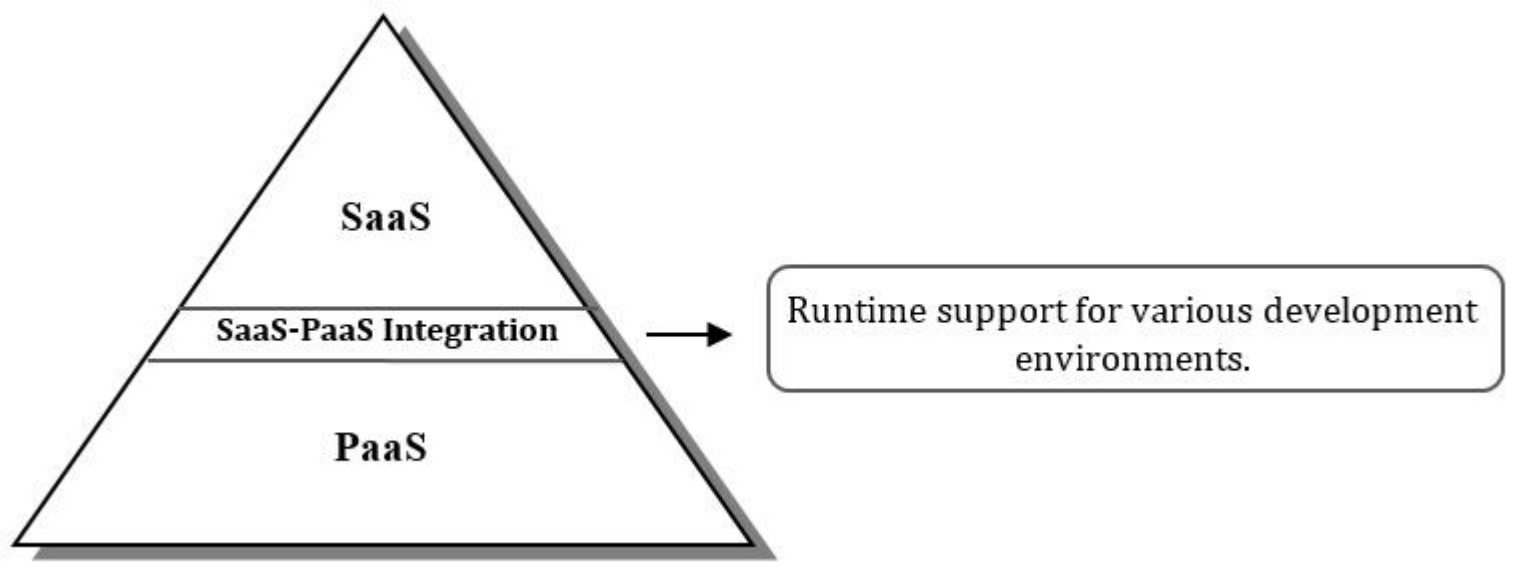
Software as a Service (SaaS)

- SaaS was the first to mature among the cloud services.
- The CRM (Customer Relationship Management) package of Salesforce.com gained popularity among enterprises from early 2000.
- Example of popular SaaS –
  - Google Apps
  - Microsoft Office 365

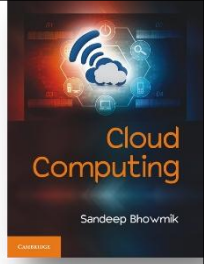


Software as a Service (SaaS)

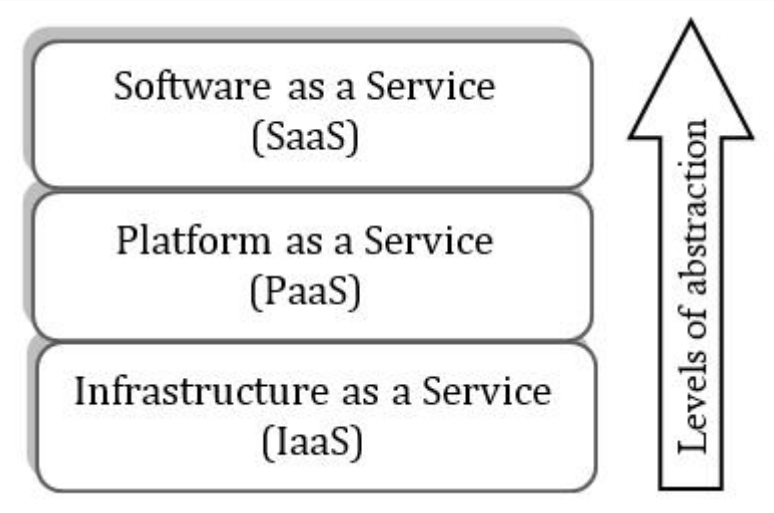
- **SaaS-PaaS Integration**



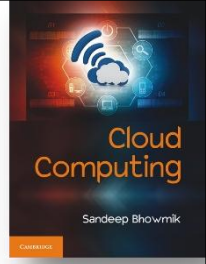
- PaaS and SaaS integration is supported by the topmost component of PaaS stack, which is runtime.



Service Abstraction

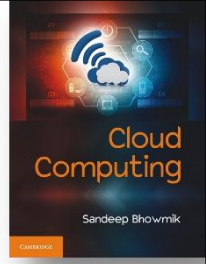


- The level of service abstraction raises when moved from IaaS towards SaaS.



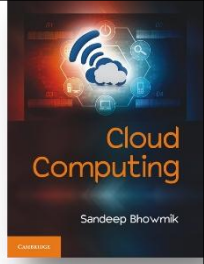
Service Abstraction

- At the IaaS level, consumers have full freedom of building computing infrastructure (in virtual mode) from a set of available options.
- In PaaS level, the abstraction intensifies. The underlying virtual infrastructure of IaaS level remains hidden from the consumers.
- At SaaS level, consumers remain unaware even about an application's technology.

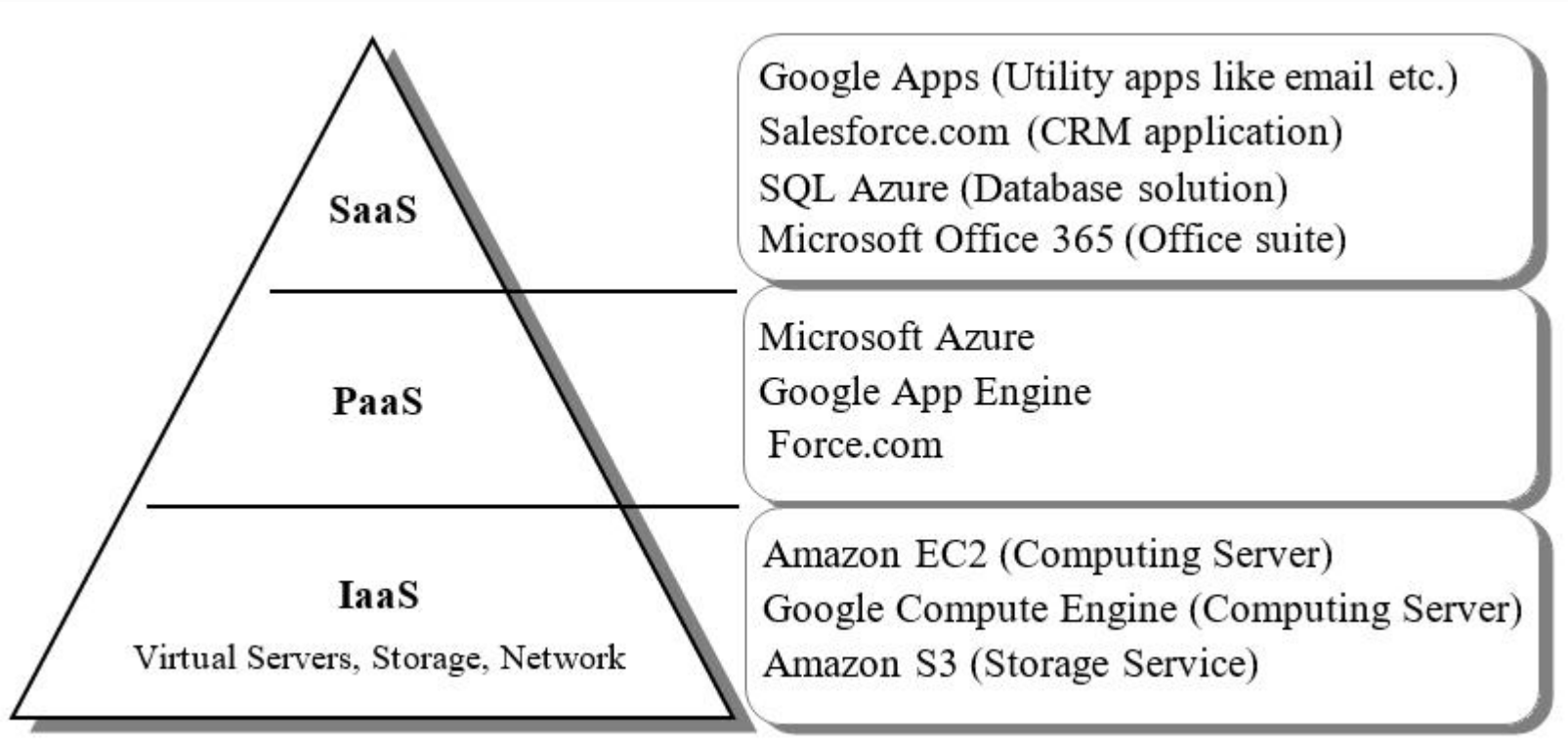


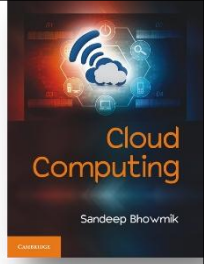
The SPI model

- The three service models **S**aaS-**P**aaS-**I**aaS together is referred as the SPI model.
- Each service layer is empowered by the capabilities of the service layer(s) beneath it.
- PaaS is empowered by IaaS layer, and SaaS is empowered with the potentials of underlying IaaS and PaaS layers.



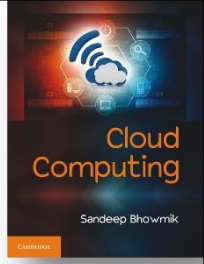
The SPI model





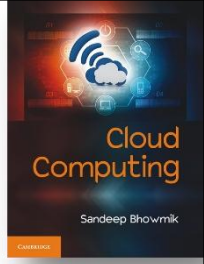
Traditional System vs. Cloud System Model

- In traditional computing environment, anyone working at some layer of this stack will have to take the responsibility of managing all the underlying layers.
- In cloud computing, users' responsibility (of managing the environment) decreases as they move from IaaS towards SaaS.
- The degree of abstraction is highest at the SaaS level, where consumers can use applications without bothering about anything underlying.



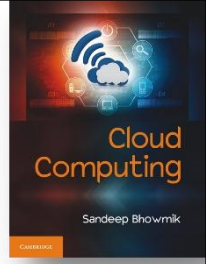
Hosted Application vs. SaaS

- All applications delivered using web-services are not SaaS.
- A hosted application can only be treated as SaaS, if that holds the basic attributes of cloud computing model.



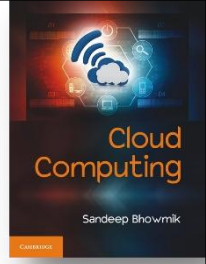
Application Service Provider

- Application Service Provider (ASP) of the 1990s was the first attempt for Internet-delivery of software, but comparatively that was more similar to the traditional application model.
- In ASP model applications are also delivered through web.
- But, it can't be termed as software-as-a-service offering.
- SaaS extends the idea from the ASP model.



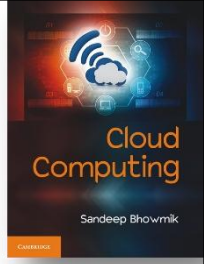
ASP model vs. SaaS model

	Application Service Provider (ASP)	Software as a Service (SaaS)
Application development	ASPs are generally not the developer of the applications they host.	In SaaS model the providers are the developer of the applications.
Deployment and management	ASPs deploy and manage applications.	Service providers of the application themselves manages everything.
Multi-tenancy	One deployment can be used by a single consumer or organization only. Multi-tenancy is not supported.	Applications are designed to be used in multi-tenant environment.
Scalability	Applications can't grow or shrink with demand. They are not scalable.	Delivers scalable applications, which can grow or shrink with changing demand.



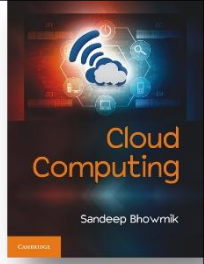
ASP model vs. SaaS model

	Application Service Provider (ASP)	Software as a Service (SaaS)
Customization	ASPs are dependent on the commercial owner of application for any customization.	Providers are the owner of the applications, and hence customization is easier.
Domain knowledge	ASPs may not have much domain knowledge of application they host.	SaaS providers are specialists in their respective application domains.
Economy of cost	Incur less cost than the traditional on-premises application model.	More economic because of the multi-tenancy and scaling ability.



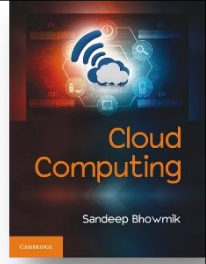
Other Category of Cloud Services

- Apart from the three primary cloud services, computing vendors deliver many specialized cloud services.
- These are considered as special services under the three primary cloud services.
- All these new "<Something> as a Service" fall into the category of XaaS (pronounced ‘zass’).



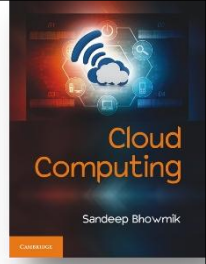
Other Category of Cloud Services

- Security Management-as-a-Service
- Identity Management-as-a-Service (IDaaS)
- Storage-as-a-Service
- Database-as-a-Service
- Backup-as-a-Service (BaaS)
- Compliance-as-a-Service
- Desktop-as-a-Service
- Monitoring-as-a-Service



Open Cloud Services

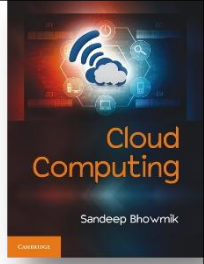
- During its initial days, all cloud computing services were proprietary.
- Services ware developed and offered by companies like Salesforce, Amazon , Google, Microsoft.
- As time passed, open source communities came up with their own solutions.
- Open source cloud community generally focuses on private cloud computing arena.



Open Cloud Services

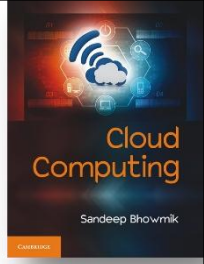
- Different open source technologies have mostly powered these open cloud services.

- Few open-source cloud solutions –
  - Eucalyptus - an open source Infrastructure as a Service (IaaS) facility for building private or hybrid cloud computing environment. It is a linux-based development.
  - OpenNebula - an open source Infrastructure as a Service (IaaS) implementation for building public, private and hybrid clouds.
  - Nebula – an open source cloud computing platform that was developed by NASA.



Open Cloud Services

- Nimbus - an open source IaaS cloud solution compatible with Amazon's cloud services.
- OpenStack - a free and open-source IaaS solution. In July 2010, U.S. based IaaS cloud service provider Rackspace.com and NASA jointly launched it.
- Apache CloudStack - an open source IaaS cloud solution.
- Enomaly ECP – an IaaS facility.



Thank You